

Lab 1

Exploratory Data Analysis and Descriptive Statistics

Explore annual baseball statistics and NYC taxi trips using Python. Formulate questions, document cleaning decisions, summarize distributions, examine sampling variability, and apply Bayes' theorem. Submit your code and interpretations in a reproducible notebook.

Learning objectives

- Ask questions that match the observational unit, variables, and time coverage of a dataset.
- Compute and interpret measures of center and spread; compare density and cumulative probability.
- Explain sampling variability and use conditional probabilities in a simple classification scenario.

Prerequisites: Python 3, Jupyter Notebook or equivalent, pandas, NumPy, Matplotlib, and SciPy. Use random seed 42 for all sampling and report package versions.

Datasets

Baseball data provided with this lab

Use data.csv. It contains 155 annual aggregate records from 1871 to 2025 and 30 columns. Each row describes a year, not an individual player. Use BA (batting average, a unitless proportion) as the main variable. Year identifies the year; G is the reported game count. R and R/G contain identical values in this file and must not be treated as separate measures.

Use years through 2024 for the main analysis. The 2025 row has a lower game count and may be an incomplete-season snapshot. The supplied file does not establish its extraction date or original provenance; acknowledge that limitation rather than claiming historical accuracy. Give each retained year equal weight and do not interpret the mean of annual BA values as a pooled player batting average.

NYC taxi data downloaded by students

Use the official 2024 Green Taxi Trip Data dataset. Each row is a trip. Download its CSV using Export on the dataset page. Preserve the downloaded file and record the dataset title, URL, and download date.

[2024 Green Taxi Trip Data and column descriptions](#)

[NYC Open Data portal supplied with the original lab](#)

Load the CSV, retain trips with pickup dates in 2024, then randomly select 10,000 rows without replacement using random_state=42. Save this raw sample as taxi_sample.csv before cleaning. Do not substitute the first 10,000 rows for a random sample. Use fare_amount (US dollars) as the main variable; it is distinct from total_amount.

1 Asking good questions

Write 2–3 exploratory questions for each dataset. Identify the variables needed and explain why the available rows can answer each question. Include at least one question about a distribution and one about an association or comparison.

Examples: How does the distribution of annual BA differ before and after 1950? How does taxi fare vary with trip distance? Avoid claims about individual baseball players or long-term taxi fare trends that these data cannot support.

2 Data loading and cleaning

- Load both CSVs and show their shapes, column names, data types, and representative rows. Record the taxi sampling procedure and the number of rows before and after each cleaning step.
- Convert numeric columns explicitly where necessary. In baseball data, treat the marker "--" as missing; RBI contains this marker. Summarize missingness by column. Do not replace unknown historical values with zero or drop rows solely because unrelated columns are missing.
- Parse taxi pickup and drop-off timestamps, using the names in the dataset dictionary. Inspect missing or invalid timestamps, nonpositive durations, negative or zero fares, duplicate records, and unusual distances or fares. Explain which records you retain or exclude and why; identical records are not automatically proven duplicate trips.
- For the main taxi fare analysis, retain finite, strictly positive fare_amount values and explain that this restricts the analysis to positive-fare trips. Do not resample to replace records removed during cleaning. Report the final sample size.
- For each main variable, identify potential outliers using the $1.5 \times \text{IQR}$ rule and inspect them. A flagged value is not necessarily an error. Justify any removal and report its effect on the analysis.

3 Descriptive statistics

For BA and fare_amount separately, report the count of retained observations and calculate:

- Arithmetic mean, median, and geometric mean.
- Sample variance and sample standard deviation, using $\text{ddof}=1$.
- Five-number summary: minimum, first quartile, median, third quartile, and maximum. Use linear interpolation for quartiles and state this convention.

Use the same retained observations for comparable statistics. Check that every value used in a geometric mean is strictly positive; do not shift values or silently remove additional records to make it calculable. Explain whether a multiplicative interpretation is useful for the selected variable.

Interpret the results in context and with units. Explain why the median is generally more robust to extreme values than the mean. State that variances have squared units and standard deviations retain the original units.

4 Density and cumulative probability

For each main variable, create a normalized histogram as an estimate of the probability density function and an empirical cumulative distribution function (ECDF). Use aligned panels with the same horizontal range, or overlay them with clearly labeled separate vertical axes.

- Label the histogram vertical axis Density and the ECDF vertical axis Cumulative probability. Give each plot a title, variable name, units, and sample size. State your histogram bin choice.
- Check that the total area of the histogram bars is approximately 1 and that the ECDF reaches 1. Density height is not itself a probability and can exceed 1.
- Explain that integrating density gives cumulative probability. For a histogram, add bar height multiplied by bin width; simply adding heights is incorrect. The ECDF comes directly from sorted observations and need not exactly equal the integrated histogram estimate.
- Read and interpret one probability or percentile from each ECDF. Discuss which features are easier to see in the histogram, such as modes, and which are easier to read from the ECDF, such as proportions below a threshold.

5 Sampling variability experiment

Perform the following experiment separately for the retained BA and fare_amount values. Treat each cleaned dataset as an empirical population for this simulation.

- Using random seed 42, independently draw 1,000 samples with replacement of size 5 and compute each sample mean. Repeat with sample size 30. Sampling with replacement allows the same observation to appear more than once within a sample.
- Plot the two distributions of sample means on comparable axes. Report their means, sample variances, and sample standard deviations using `ddof=1`.
- Compare the simulated variances with the empirical population variance divided by the sample size. For this benchmark, calculate the variance of the retained source values with `ddof=0`. Explain why this differs from the sample-variance convention used in Task 3.
- Discuss whether larger samples yield less variable means and whether the simulation agrees approximately with the benchmark. Monte Carlo results need not match it exactly.

Reflection: Explain why this simulation does not establish independence in real historical years or taxi trips. Discuss temporal changes, related observations, and the limits of generalizing beyond the supplied data. Averaging equal-sized folds of a fixed dataset reproduces its original mean; it does not by itself improve that estimate's precision.

6 Bayes theorem

Use the baseball records retained for analysis through 2024. Imagine selecting one of these years uniformly at random. Define event A as Year at least 1950 and event B as BA at least 0.260. Here, not A means Year before 1950.

- Construct a two-by-two count table for A versus not A and B versus not B. Show row totals, column totals, and the overall total.
- Calculate $P(A)$, $P(\text{not } A)$, $P(B \text{ given } A)$, and $P(B \text{ given not } A)$. Show the counts and denominators used.
- Apply Bayes' theorem to calculate $P(A \text{ given } B)$. Show your substitution and calculation, including the total-probability calculation for $P(B)$.
- Verify the result by directly dividing the number of years satisfying both A and B by the number satisfying B. Check that the denominators are nonzero and that the two methods agree.
- Interpret the posterior probability in a sentence. Explain the distinction between $P(A \text{ given } B)$ and $P(B \text{ given } A)$, and how the result could inform a simple classification of a randomly selected year. Do not make causal claims or claim that this establishes predictive accuracy on new data.

Deliverables

- One Jupyter notebook (.ipynb) with questions, executable code, outputs, plots, and short written interpretations for all six tasks. Restart the kernel and run all cells in order before submission.
- A README summarizing findings in plain English, data sources and limitations, cleaning and sampling choices, package versions, and instructions for running the notebook. Include the taxi download link and date.
- Include taxi_sample.csv, or provide reproducible download and sampling instructions with the retained source-file details. Use relative file paths. Identify the supplied baseball CSV as course-provided data with unverified original provenance.

Evaluation criteria

Criterion	Points
Exploratory questions and alignment with the data	10
Correct loading, cleaning, statistics, and sampling code	30
Accurate and clearly labeled visualizations	20
Interpretations, limitations, and reproducibility	25
Correct Bayes calculation, verification, and interpretation	15
Total	100

Dataset links checked on September 16, 2026. Consult the taxi dataset page for field definitions and current download options.